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Highlight Review

The Many Ways for Reversible Polymerization of 1,2-Dithiolanes

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Abstract

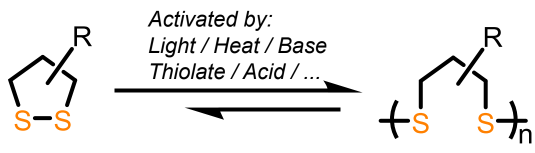
The reversible ring-opening polymerization (ROP) of 1,2-dithiolanes offers a versatile dynamic chemistry platform to design materials inherent in dynamic properties. Due to the dynamic covalent nature of disulfide bonds, the ROP of 1,2-dithiolanes can be triggered by several methodologies to control the architectures and self-assembly of the resulting polymers. This Highlight Review focuses on the recent advances in developing the methodologies to make and break (recycle) the polymers of 1,2-dithiolanes.

Keywords: 1,2-Dithiolanes | Reversible polymerization | Dynamic chemistry

Introduction

The disulfide bond, one of the most common dynamic covalent bonds, has been widely used in designing dynamic systems and materials. The moderate bond energy (60 kcal/mol)[1] of disulfide bonds endows them both robustness and dynamicity, for example, they can readily exchange with thiols rapidly even under mild conditions.[2,3] This feature makes disulfides very useful in designing dynamic materials, especially polymers, because their simple structure can be readily introduced.[4] Using disulfide bonds as the reversible linkages in the sidechain or mainchain backbone of polymers can result in dynamic covalent polymers that exhibit self-healing properties, reconfigurability, and stimuli-responsive ability,[5–10] bringing many opportunities in the fabrication of smart materials.

Recently, a series of poly(disulfide)s made from the ring-opening polymerization (ROP) of cyclic 1,2-dithiolanes (e.g. thioctic acid, TA) have attracted much attention because it bridges dynamic covalent chemistry with ROP. The cyclic disulfides exhibit unique capabilities of reversible polymerization owing to the ring strain of the five-membered 1,2-dithiolanes,[11] generating poly(disulfide)s easily depolymerized and even recycled. Remarkable efforts have been made in developing poly(1,2-dithiolane)s for diverse applications such as surface polymerization,[12,13] cell-penetration,[13] protein-polymer conjugates,[14] gels,[15–17] elastomers,[18,19] self-healing materials,[18–23] supramolecular assembled materials,[23,24] adhesives,[18,25] clusteroluminescence,[26] and chemically recyclable materials.[22–24,27–30]



Scheme 1. Schematic illustration of the reversible ROP of 1,2-dithiolanes.

These rising research interests have motivated us to give a highlight summary on the recent advances in the polymerization methodologies of 1,2-dithiolanes. It should be noted that here we do not aim at giving a comprehensive overview for this field, but at highlighting the most robust and versatile strategies towards the controllable preparation and chemical recycling of poly(1,2-dithiolane)s. Several ways to activate disulfide bonds will be discussed, including light (photopolymerization), heat (thermal polymerization), base/thiolate (anionic polymerization), and acid (cationic polymerization) (Scheme 1).

Polymerization Methodologies

The first example of the synthesis of poly(1,2-dithiolane)s was reported by Calvin et al. by photopolymerization in 1954 (Figure 1).[31] Trimethylene disulfides were irradiated by UV light at low temperature (−196 °C) in rigid solvent, where the disulfide bonds broke into diradicals and remained stable, followed by the production of polymers by intermolecular connection after warming a few degrees. More detailed investigation on the photo-induced ROP of 1,2-dithiolanes was carried out by our group.[20] TA monomers were heated to a solvent-free molten state, after treating with visible light (420 nm), quantitative conversion of monomer to polymer was achieved and resulted in cyclic poly(TA) mediated with a diradical mechanism. The advantage of this method is that cyclic disulfides are intrinsic chromophores, thus enabling an initiator-free photopolymerization methodology.

Thiols can be oxidized into disulfides,[32] and poly(disulfide)s can also be made by oxidation polymerization of dithiols. The synthesis of poly(1,2-dithiolane)s by oxidation method originated from an unexpected discovery by Thomas and Reed in 1956 (Figure 1).[33] When exploring the synthesis of TA by oxidating 6,8-dimercaptooctanoic acid,[34] they observed that sticky and colorless polymers were produced as by-products, indicating that partial monomers were connected by intermolecular oxidation. It was also reported that poly(TA) was obtained by heating TA monomer, although detailed characteristics were not presented.

The thermal polymerization of 1,2-dithiolanes was investigated by Endo et al. in 2006 (Figure 1).[35] On the basis of synthesis and characterization of poly(1,2-dithiane)s,[36] they prepared high-molecular-weight copolymers (up to 550 kDa) of 1,2-dithianes and 1,2-dithiolanes through initiator-free one-pot thermal copolymerization. The structure was inferred to be cyclic mainly because: i) In the ^{1}H NMR spectrum, the peaks of thiols were absent, which should have been present at the end of linear chains; ii) Dynamic viscoelasticity measurement showed a rubber plateau when the temperature was over 0 °C and the region became smaller with the decrease of the frequency, suggesting that there were intermolecular entanglements due to the topologically linked cyclic structure. All these experimental observations implied the formation of polycatenane-like topology of the resulting polymers, despite the lack of direct evidence proving the hypothesis due to the difficult characterization of these dynamic and catenated systems.

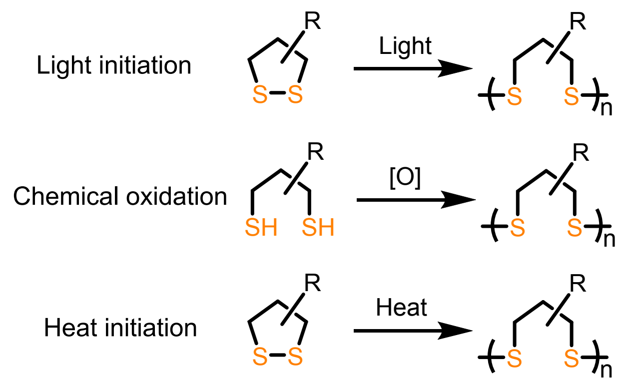


Figure 1. Schematic illustration of the ROP of 1,2-dithiolanes initiated by light, chemical oxidation and heat.

Taking advantage of thermal polymerization of 1,2-dithiolanes, our group developed a series of dynamic polymers based on TA and its derivatives through one-pot synthesis.[18,19,21] By melting TA powder at 70 °C and adding cross-linkers diisopropenylbenzene (DIB) and iron(III) ions, a free-standing network containing dynamic covalent bonds, H-bonds and coordination bonds was obtained (Figure 2A).[18] Benefiting from the supramolecular linkages, the resulting supramolecular polymers were highly stretchable and showed good performance in self-healing ability. The iron(III) ions was able to enhance the strength of materials as a result of the formation of iron(III)-carboxylate complexes. Furthermore, the molten solvent-free environment enabled the generation of secondary ionic clusters (Figure 2B), leading to significant increasement of Young's modulus from 81 KPa to 5.1 MPa while maintaining stretchability and self-healing properties.[19] Although the dynamic poly(TA) network is hard to apply in robust high-strength materials due to the weak internal interactions, the side chain modification engineering strategies may be able to change the properties of materials. Our group further synthesized thioctic acylhydrazine (TAH) monomers and obtained poly(TAH) via thermal solvent-free ROP (Figure 2C).[21] Each hydrazide on the side chain was able to form six intermolecular H-bonds, this unique reticular H-bonding cross-link endowed the poly(TA-H) network remarkably strengthened Young's modulus (up to 340.3 MPa). Overall, these examples show the easy (one-pot) and green (atomically economic) synthesis of poly(1,2-dithiolane)s, as well as their applications in self-healing materials.

Although the methodologies of ROP of 1,2-dithiolanes have been developed for decades, the living polymerization method has only been investigated in recent years,[12–14,16,37] that is, utilizing active thiolates to trigger the cyclic disulfides, followed by nucleophilic attack on other monomers. The methodology of living anionic polymerization enables the controllability of degree of polymerization, molecular weight distribution and topology, etc. The first example was reported by Matile et al. for self-organizing surface-initiated polymerization in 2011 (Figure 3A).[12] A symmetric 1,2-dithiolane (asparagusic acid,

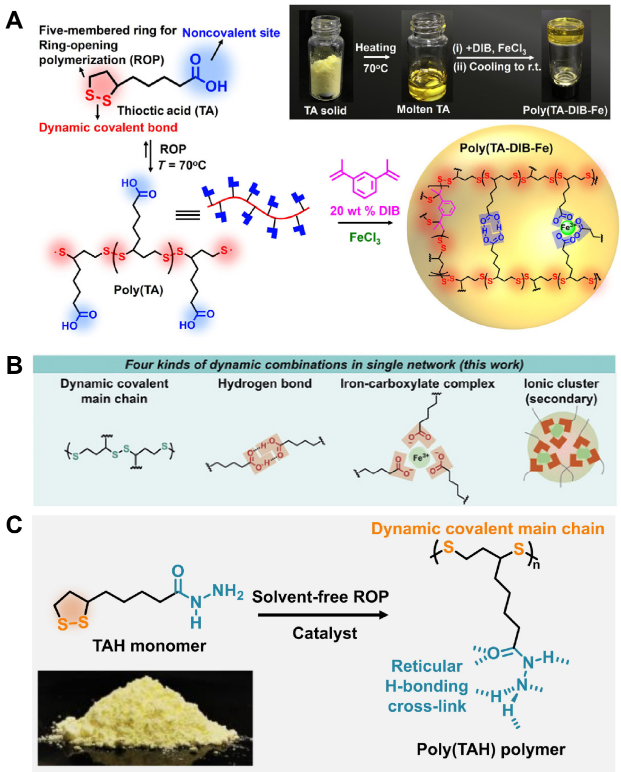


Figure 2. Supramolecular polymers of TA and its derivatives obtained by thermal polymerization. (A) Poly(TA) network linked by DIB and iron(III) ions. Reproduced with permission from ref 18. Copyright 2018 American Association for the Advancement of Science. (B) The four forces existing in the poly(TA) network after increasing the content of iron(III) ions, including ionic clusters. Reproduced with permission from ref 19. Copyright 2020 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim. (C) The ROP of TAH and the formation of reticular H-bonding cross-link. Reproduced with permission from ref 21. Copyright 2022 American Association for the Advancement of Science.

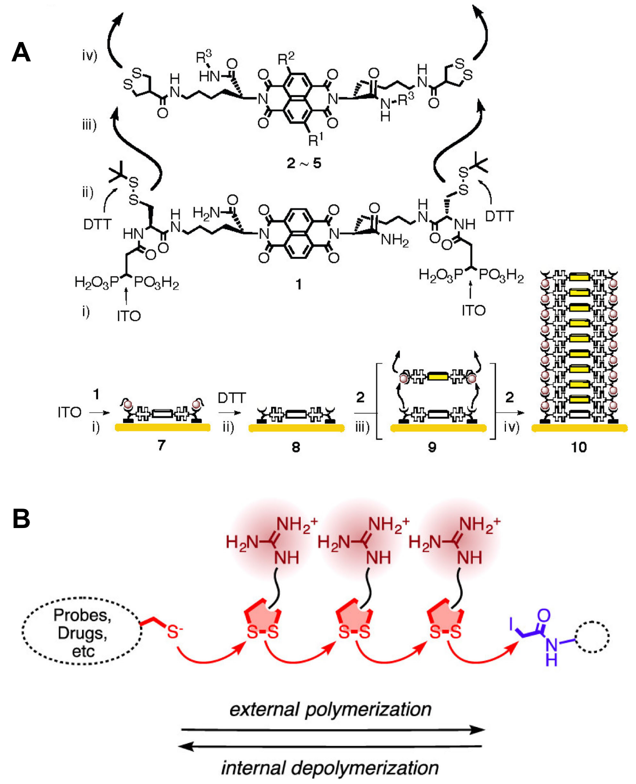


Figure 3. (A) Anionic living polymerization of 1,2-dithiolanes for self-organizing surface-initiated polymerization. Reproduced with permission from ref 12. Copyright 2011 American Chemical Society. (B) The substrate-initiated synthesis of cell-penetrating poly(1,2-dithiolane)s by anionic polymerization. Reproduced with permission from ref 13. Copyright 2013 American Chemical Society.

AA) was designed to be the end of a building block consisting of aromatic-stacking and H-bonds. With the *in-situ* generation active thiolate initiators by (S,S)-dithiothreitol on the substrate, the cyclic disulfide bonds in the AA units were triggered and occurred ROP accompanied by supramolecular self-assembly. As a result, highly controlled layered polymer was obtained.

Furthermore, in 2013, the Matile group further applied this strategy on preparing cell-penetrating poly(disulfide)s in aqueous/MeOH solution (Figure 3B).[13] The 1,2-dithiolane monomers were modified with guanidinium cations on the side chains for water-solubility. Probes or drugs containing thiols can be used as the substrate to initiate the ROP of water-soluble 1,2-dithiolanes, which significantly enhanced the cell-penetrating ability of the drugs. The most important feature of such a system is the capability to depolymerize into monomeric molecules by treating with dithiothreitol (DTT), providing a noninvasive, nontoxic way to deliver original substrates.

In 2015, Waymouth et al. confined the thiolate-initiated ROP of 1,2-dithiolanes in an amphiphilic block polymers for constructing structurally dynamic hydrogels (Figure 4A).[15] They designed water-soluble "ABA" triblock copolymers containing pendant 1,2-dithiolanes, which were capable of generating hydrogels upon treating with dithiols to connect the pendant 1,2-dithiolanes. The hydrogels exhibited dynamic behaviors due to the reversible ROP of pendant 1,2-dithiolanes. The hydrogel began to flow when the vial was inversed, while the hydrogel whose free thiols capped by maleimides did not flow; this difference strongly implicated that the thiols-coursed rapid and reversible ROP of pendant 1,2-dithiolanes resulted in the dynamic behaviors. Based on this system, the same group further investigated the thermodynamic properties of 1,2-dithiolanes in 2017 (Figure 4B).[16] They prepared two model monomers TA methyl ester (TA-Me) and methyl asparagusic acid methyl ester (MAA-Me). The thermodynamic parameter results revealed that the ROP of 1,2-dithiolanes was driven by the enthalpy originating from the release of ring strain. The TA-Me exhibited much higher equilibrium constant K_{eq} and a more negative enthalpy ΔH_p than MAA-Me, indicating a larger ring strain of TA-Me compared to MAA-Me. The significant difference in thermodynamic behavior between them is supposed be attributed to the Thorp-Ingold effect,[38] that is, the extra methyl substitution in MAA-Me reduces the bond angle of the five membered ring, which benefits the depolymerization reaction to generate ring monomers.

For most polymerization methods of poly(1,2-dithiolane)s, the topologies are single and difficult to control. For example, the

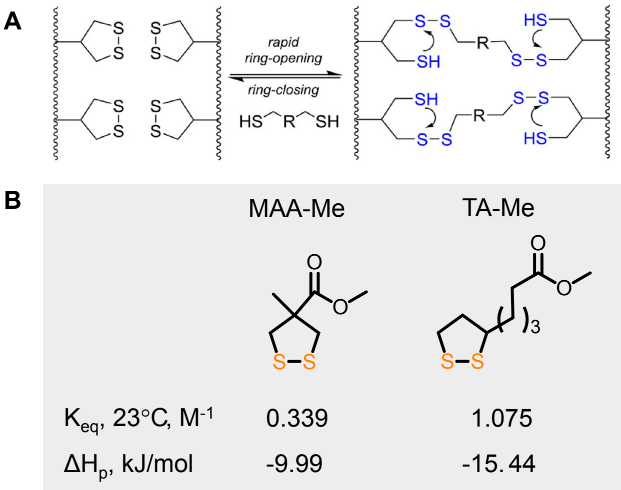


Figure 4. (A) Dynamic hydrogels based on the thiolate-initiated reversible ROP of 1,2-dithiolanes. Reproduced with permission from ref 15. Copyright 2015 American Chemical Society. (B) Thermodynamic data of MAA-Me and TA-Me reported by Waymouth et al.

photo-initiated and thermal-initiated polymerizations are considered to offer cyclic poly(1,2-dithiolate)s.[20,35] The thiolate-initiated polymerizations reported by Matile et al. and Waymouth et al. using alkyl thiols as the initiators obtained linear poly(1,2-dithiolate)s.[13,16] In 2019, Moore et al. discovered methodology to control the topology of poly(1,2-dithiolane)s through an anionic ROP approach (Figure 5A).[37] Cyclic polymers were obtained by employing aryl thiolates (e.g., Ph-SH), while the alkyl thiolates led to linear polymers. The ability of aryl thiolate to produce cyclic species was attributed to its better nucleofugality compared to alkyl ones, thus it was a good leaving group to promote the formation of rings. Both cyclic and linear polymer thermodynamic aspects were studied. In this work, benefiting from the feature of living polymerization of anionic ROP, they achieved precise control during the polymerization such as topologies, conversions, kinetics, molecular weights and so on, which revealed the anionic ROP of 1,2-dithiolanes in great detail.

The successful propagation of poly(1,2-dithiolane)s in aqueous solution indicated the application prospect in biomaterials.[12,13,39–41] However, the high equilibrium monomer concentration ($[M]_{eq}$) at room temperature limited the conversion of monomer to polymer. An approach to carry out the polymerization in frozen environment reported by Lu et al. turned out to be feasible to promote the ROP of 1,2-dithiolanes in aqueous phase (Figure 5B).[14] The reduced temperature attenuated the entropy penalty, which greatly favored the thermodynamics. At the same time, the frozen systems made the solutes concentrated as a result of the solvent expulsion through forming ice crystals, leading to enhanced kinetics. Utilizing the inherent thiol groups in proteins, high conversion and rapid preparation of protein-poly(disulfide)s conjugates were achieved by *in-situ* polymerization from protein surface-tethered initiators. This strategy also allowed the reversible depolymerization under mild conditions to regenerate original proteins. These features enabled the poly(1,2-dithiolane)s to graft and release proteins.

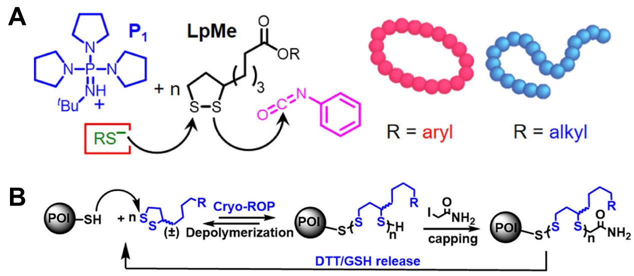


Figure 5. (A) Architecture control of poly(1,2-dithiolane)s by anionic ROP using aryl or alkyl thiolate initiators. Reproduced with permission from ref 37. Copyright 2019 American Chemical Society. (B) Synthesis of protein-poly(disulfide)s conjugates via *in situ* polymerization of 1,2-dithiolanes from protein surface-tethered initiators in frozen environment. Reproduced with permission from ref 14. Copyright 2020 American Chemical Society.

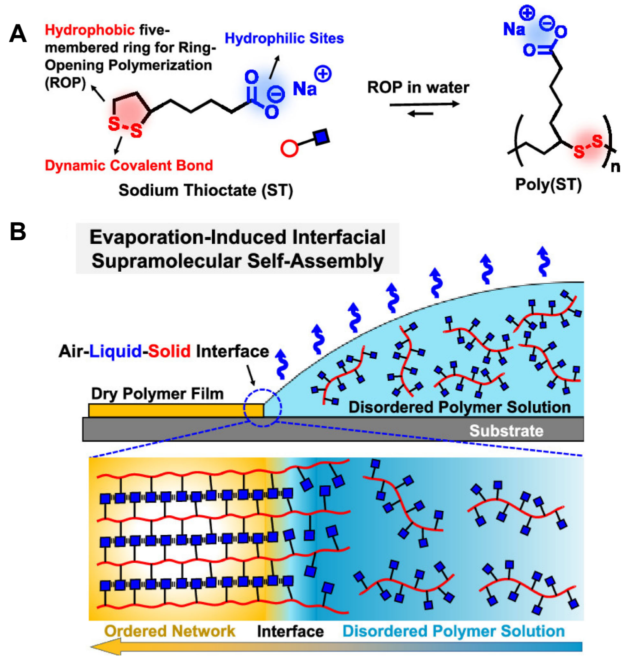


Figure 6. (A) Molecular illustration of the amphiphilic structure of ST monomer and poly(ST). (B) Schematic illustration of the evaporation-induced interfacial self-assembly process of ST and the resulting highly-ordered poly(ST) with layered structure. Reproduced with permission from ref 23. Copyright 2019 American Chemical Society.

For most of the examples of poly(1,2-dithiolane)s, the resulting polymers are amorphous due to the lack of orientation selection during the polymerization process. Utilizing hydrophilic and hydrophobic forces as well as interfacial self-assembly induced by evaporation, our group explored the highly-ordered ROP of 1,2-thiolanes in aqueous phase to assemble polymers with layered structure (Figure 6).[23] By deprotonating the carbonyl groups of TA, high water soluble amphiphilic sodium thioctate (ST) was obtained. This slow thermodynamic controlled process enabled the spatially confined growth of dynamic covalent polymers accompanied by supramolecular ionic cross-linking, leading to the formation of highly-ordered layered architectures. The unique ordered structure endowed the poly(ST) materials humidity responsiveness and self-healing ability in humid environments. Moreover, the polymer could be readily depolymerized to monomers by dissolving in water. Further, following this evaporation-induced strategy, a series poly(1,2-thiolane)s with various β-sheet-like H-bonds terminal groups were prepared.[24] The layer space could be precisely controlled by different lengths of side chains. The layered structure and terminal hydrophilic groups promoted the formation of interlayer water channels, enabling the assembled ordered polymer efficient ion transporting performance. These two examples demonstrated the green synthesis of highly-ordered disulfide-mediated polyelectrolytes through simple one-pot evaporation in water, which provided a convenient, low-cost, and pollution-free way for synthesizing layered structure polymers.

Despite the thiolate-initiated anionic ROP being investigated widely and controlled precisely, there are still several remaining aspects limiting its practical application, especially in materials. The thiolates are easily oxidized by air so strict inert conditions are required. The basic condition also makes it unsuitable for monomers with active protons. Therefore, a simple and protection-free methodology that can efficiently trigger the ROP of 1,2-dithiolanes is highly desirable. Recently, our group reported the first example of cationic ROP of 1,2-disulfides through an acid-catalyzed pathway (Figure 7A).[30] The polymerization can be initiated by adding catalytic amounts of acids (e.g., trifluoroacetic acid, TFA) into the monomers in air without any protection, leading to the gelation of monomer solution to a free-standing gel (Figure 7B), resulting in poly(1,2-dithiolane)s with high molecular weight (over 1000 kDa) within few minutes (Figure 7C). The mechanism is considered to be cationic polymerization initiated by sulfonium ion protonated by H^+, which was supported by the cocatalyst effect of non-coordinating anions (e.g., hexafluorophosphates, PF_6^-). The addition of PF_6^- stabilized the sulfonium active sites and enhanced the conversions. In addition, acid catalyst was also able to catalyze the depolymerization of polymer efficiently and obtained recycled monomer.

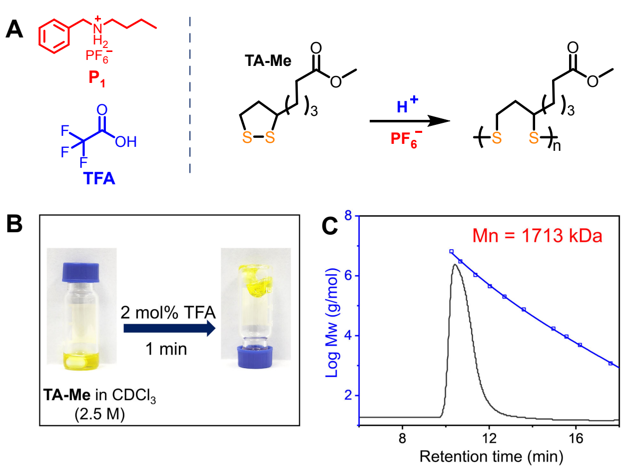


Figure 7. (A) Molecular illustration of the acid-catalyzed cationic polymerization of 1,2-dithiolanes. (B) Photographs of TA-Me solution and rapid gelation after adding acid catalyst. (C) GPC analysis of poly(TA-Me). Reproduced with permission from ref 30. Copyright 2023 Wiley-VCH GmbH.

Depolymerization Methodologies

The inherent dynamics of disulfide bonds endows 1,2-dithiolanes the reversible polymerization ability, i.e., the polymers can be depolymerized under mild conditions due to the low ceiling temperature (T_c).[30,37] This feature benefits development of disulfide-mediated degradable or recyclable polymers. The progress of depolymerization methodology is often accompanied by the development of polymerization methodology. The depolymerization reaction can be triggered by external conditions (e.g., alkaline, UV-light, heat and thiols)[13–15,20,27,30,35] or changing the thermodynamics of the polymerization reaction to reverse the balance via adjusting the temperature or monomer concentration.[30,37]

Reed et al. reported that poly(TA) obtained by oxidation could depolymerize to TA monomers in dilute sodium hydroxide solutions.[33] The mechanism was proposed to be anionic exchange or displacement reaction of S^- with disulfide linkages due to the fact that i) The depolymerization rate was correlated with the degree of alkalinity; ii) addition of thiols was able to promote the depolymerization; iii) The depolymerization reaction was completely inhibited after the addition of thiol inhibitors until the inhibitors were quenched. In 2021, our group reported the detailed depolymerization and chemical recycling of poly(TA) polymers in a dually closed-loop manner via base-catalyzed method (Figure 8).[27] The TA-mental networks were readily depolymerized upon dissolving in sodium hydroxide aqueous solution. After purification from the mixture by simply filtering, recycled TA monomers with virgin quality were obtained and could be used to prepare poly(TA) polymers. This process could be cycled multiple times without damaging polymer properties, showing the robust strategy on chemical recycling of poly(1,2-dithiolane)s.

UV light was proved to be able to activate the bulk 1,2-dithiolane monomers, resulting in high-yield polymers.[20] By placing the polymers in an excess solvent (i.e., diluting the monomer concentration), Endo et al. reported that the molecular weights of polymers were decreased remarkably at the initial stage, followed by remaining almost constant for a prolonged time, suggesting that the interlocked cyclic polymers were cut

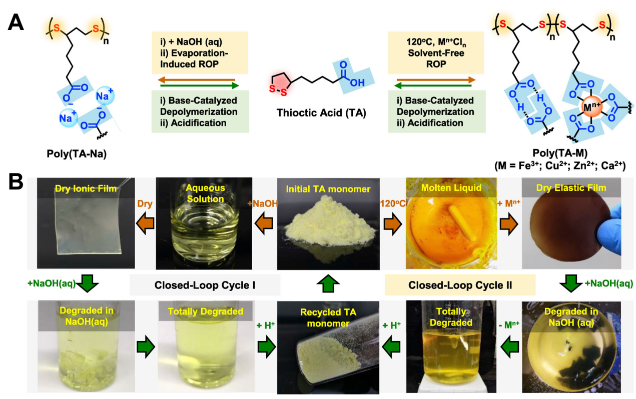


Figure 8. (A) Molecular structures of TA monomer and the resulting poly(TA). (B) Photographs of the dual closed-loop chemical recycling of TA in sodium hydroxide aqueous solution. Reproduced with permission from ref 27. Copyright 2021 Elsevier Inc.

and degraded via the irradiation of UV light.[35] For poly(1,2-dithiolate)s in bulk phase, our group recently demonstrated that heat was able to activate the polymers and regenerate monomers,[20] although the yield was relatively low, which might be attributed to the limitation of thermodynamics due to the ultrahigh monomer concentration.

The reversible thiol-disulfide exchange reactions turned out to be also applicable for thiol-initiated depolymerization for poly(1,2-dithiolane)s. Upon introducing thiols (e.g., DTT), both cell-penetrating poly(disulfide)s reported by Matile et al.[13] and protein-poly(disulfide)s conjugates reported by Lu et al.[14] were depolymerized and the original substrates were released at the same time. In Waymouth's work,[15] the free thiols in disulfide-linked hydrogels made the gels flow as a result of thiol-initiated reversible ROP reaction with poly(1,2-dithiolane)s, while the maleimide capped counterparts did not flow since the free thiols were quenched. Recently, we also reported that the treatment with thiols would significantly reduce the molecular weights of poly(1,2-dithiolane)s at room temperature.[30] These results reveal that thiols can effectively depolymerize poly(1,2-dithiolane)s under mild conditions.

Adjusting thermodynamic equilibrium is also a common way to depolymerize polymers and achieve chemical recycling to monomers.[42,43] The polymerization enthalpy ΔH_p of 1,2-dithiolanes was about $-18\,\mathrm{kJ\,mol^{-1}}$,[30] thus the depolymerization process is an endothermic reaction. The low ceiling temperature enables poly(1,2-dithiolane)s to depolymerize under modified polymerization conditions that simply raise temperature or reduce monomer concentration. In Moore's work, they diluted the polymer solution from 3.5 M to 1.0 M and raised the temperature from 25 °C to 65 °C, the original initiator thiolates for polymerization became a catalyst for depolymerization, and most polymers were recycled to monomers in an hour (Figure 9A).[37] Similarly, in our acid-catalyzed method,[30] the diluted solution of polymers showed ultrafast depolymerization behavior at room temperature after introducing trifluoromethanesulfonic acid (TfOH), this second-to-minute scale process was monitored by UV/Vis spectroscopy (Figure 9B). Increasing the temperature to 61 °C even achieved quantitative depolymerization within 5 minutes. Further, we applied this new methodology to protection-free synthesis and chemical recycling of bulky cross-linked poly(TA)s, showing practical application in recyclable materials.

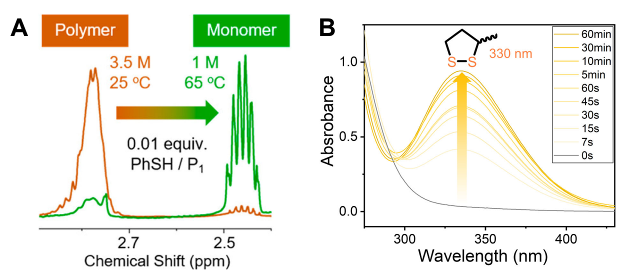


Figure 9. (A) ^{1}H NMR spectra of thiolates-catalyzed depolymerization of poly(1,2-dithiolane)s in 65 °C. Reproduced with permission from ref 37. Copyright 2019 American Chemical Society. (B) UV/Vis spectra of the monomer regeneration during the acid-catalyzed depolymerization process at room temperature. Reproduced with permission from ref 30. Copyright 2023 Wiley-VCH GmbH.

Conclusion

In summary, we gave a concise overview on the development of (de)polymerization methodologies of 1,2-dithiolanes. Benefiting from the inherent dynamics of disulfide bonds, the reversible polymerization can be achieved by many methods. These advances have made great contribution to the development of disulfide dynamic covalent chemistry. In the future, there is still much space to improve the existing methods, for example, except for the only living polymerization of 1,2-dithiolanes by anionic mechanism,[12,13,16,37] new living polymerization methodologies are expected. In addition, the regioselectivity of unsymmetrical monomers (e.g., TA) has not been achieved to date (i.e., the two kinds of chemically unequal sulfur atoms are randomly arranged in the polymer chains),[30,37] which may be probably solved by more precise control during the polymerization process. We anticipate that the methodologies in this field will be developed towards a more sustainable, controllable, and greener manner, which will bring many opportunities for disulfide-mediated dynamic covalent chemistry.

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